



Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 3B Further Statistics 1

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General comments

This paper allowed learners across the full range of ability to demonstrate their understanding of the specification. The early questions on discrete random variables and contingency tables were accessible, and many learners were able to make good progress through the paper as a result. Learners generally coped well with the Poisson, Geometric and Negative Binomial content, and with applying the central limit theorem. Part (b) of the final question proved more demanding, with fewer learners able to structure a complete solution.

As in previous sessions, learners are reminded of the value of showing clear method, using the language and notation given in the question for conclusions, and taking care with rounding and with the direction of hypotheses.

Comments on individual papers

Question 1

Part (a)

Most learners gained both marks in this part; where a mark was not awarded, this was usually the result of misreading the table. A small number of responses did not find the correct expected frequencies.

Part (b)

Many learners correctly gave the hypotheses in terms of association, although some responses wrote them in a different way. A minority of responses referred to independence and dependence instead, and some of these were again written in a different way. Most learners expressed the hypotheses in context using the appropriate language. Some learners expressed the hypotheses in terms of correlation or a relationship rather than association or independence, which did not gain the mark.

The calculation of the test statistic was generally done well, with learners who had an earlier incorrect value able to pick up a method mark here. Some learners did not show their method, and marks were not awarded as a result of using an incorrect value from part (a), calculator error, or early rounding leading to an inaccurate answer. A common error was to treat 6.314 as the test statistic for all the data, despite the question stating otherwise, rather than adding on the two other contributions. Some learners chose to recalculate all nine components; a small number of these then added 6.314 to their answer as well.

Almost all learners stated the degrees of freedom correctly and went on to gain the mark for the associated critical value. A few learners did not gain this mark by giving the value to three significant figures.

Most learners wrote a conclusion in context using the language given in the question. Some learners concluded that when the test statistic exceeded the critical value they should not reject H_0 , leading to an incorrect conclusion. Learners who had written the hypotheses in a different way usually often the wrong conclusion as a result. Some responses did not clearly state what the result meant in relation to the hypotheses being tested, and contradictory statements such as “reject H_0 , we have insufficient evidence to support Diogo’s belief” were seen.

Question 2

Part (a)

This part was answered well, with the most learners able to find the correct values for p and q . Most learners observed that they needed to select the value of p that was a valid probability, although some unnecessarily substituted $p = 1.8$ into an equation to find q before rejecting this solution. There was the occasional response that showed algebraic sign slips when manipulating the equation to solve for p , although this was rare.

Part (b)

In this part, most responses demonstrated a successful method by calculating $4\text{Var}(X)$ or $\text{Var}(Y)$ from a probability distribution table where $Y = 2X + 3$. Some responses, however, showed the incorrect calculation of $2\text{Var}(X)$, which did not gain the final two marks. Some responses did not always show a clear method for finding the required components of the calculation, and values were not always labelled. Other responses omitted squaring $E(X)$ when finding $\text{Var}(X)$.

Part (c)

Part (c) was found to be challenging for learners. Many identified the three cases where $X_2 - X_1 > 7$; some responses showed the construction of a possibility space diagram for X_2 and X_1 , with others doing so by comparing values of $X_1 + 7$ (written above the given probability distribution table) with X_2 . Other responses partially showed use of an exhaustive method, e.g. if $X_1 = -5$, $X_2 > 2$; if $X_1 = -3$, $X_2 > 4$; no cases with other values of X_1 . Others simply stated the cases without a method for identifying them. A number of responses incorrectly included the extra case of $X_2 = 0$ and $X_1 = -5$. Some responses identified only two of the three cases, and occasionally only one case was identified. There were a number of errors substituting the probabilities corresponding to the required values of X_1 and X_2 accurately, with some responses omitting to use independence and multiply the probabilities together; a few responses added when they should have multiplied for each case, and multiplied when they should have added to combine the three cases.

Question 3

Part (a)

This part was generally well answered, with most responses stating the appropriate Poisson distribution using $\lambda = 24$ and applying this correctly. In (ii), most learners were successful in correctly interpreting the question as $P(X > 14)$ and used their calculator appropriately to find this. Only a few responses mistakenly calculated $P(X \geq 14)$. There was the occasional error in rounding to three significant figures, leading to the loss of the final B1 or A1 mark.

Part (b)

This part was generally well answered, with responses showing good knowledge of adding Poisson distributions and achieving $\lambda = 68$, although common incorrect distributions of $Po(56)$ or $Po(60)$ were seen, leading to the loss of both marks. Learners are reminded to read the context carefully, as this required the total number of vehicles, not just electric cars.

Part (c)

This part was not as well answered as parts (a) or (b). Most responses gained the first mark for finding the probability of exactly 10 electric cars, but a number then continued with a Binomial distribution instead of converting to a Poisson distribution as the question asked. Of those that did convert to a Poisson distribution, the great majority went on to complete the question successfully.

Part (d)

This part was generally well answered, although many responses gave no context, relying instead on the standard textbook statement “ n is large, p is small”. Some learners did not distinguish between the modelling assumptions of the Poisson distribution and the conditions under which a Poisson distribution can be used to estimate a Binomial probability. Some learners phrased their responses successfully in terms of np being approximately equal to npq , but a small minority referred to “ $np < 10$ ”, which did not answer the question.

Part (e)

This part was well answered well, with learners typically applying the Poisson distribution with $\lambda = 48$ and successfully calculating a probability of 0.0395. However, a small number of responses slipped with the parameter in H_0 and H_1 , with some using the wrong symbols, such as p or x , and some writing their

hypotheses in words. The question asked for the p -value, which suggested this approach should have been used for the hypothesis test; a small number of responses could not gain all the marks for using a critical region method without the supporting probabilities stated. The conclusion for the hypothesis test could have used the word 'belief', as this was given in the question; those responses who did not take this approach usually still found the correct words in the context of 'rate', 'cars' and 'increased'. A few responses used 'number of', which did not gain a mark; the words provided in the question for the conclusion must be used. Some responses did not gain the mark by not stating a correct conclusion, for example "reject H_0 , there is insufficient evidence to support Simone's belief".

Question 4

Part (a)

Many learners demonstrated the working needed to find the value 0.4, most commonly by calculating $E(X)$ and then taking the reciprocal of their answer. A few responses did not show how they arrived at 375 or simply stated $E(X) = 2.5$ without supporting working, both of which gained no marks. The most common error was attempting to use the six groups to calculate the mean of a binomial distribution.

Part (b)

This part was usually well answered, with very few slips seen when evaluating the expected frequencies. Where errors did occur, these were usually the result of a calculation error, with the correct method for either r or s otherwise evident. Learners are reminded of the value of checking whether their answer makes sense in the context of the calculation being done.

Part (c)(i)

This was a routine goodness of fit test, with the key points required given in the question, and most responses worked through the list systematically. A substantial number of responses included 0.4 in the hypotheses, and a large minority wrote the hypotheses differently, both not gaining the mark. The calculation of degrees of freedom was generally good, although some responses did not identify the extra constraint, giving 5 instead, and some combined cells because the observed frequency of 6 was less than 5, indicating some confusion of the required technique. The test statistic was generally calculated accurately unless cells had been combined; the method mark was allowed where either r or s was less than 5. Only occasionally did responses divide by the observed frequency rather than the expected. The conclusion was usually stated correctly, except where the hypotheses had been written the other way.

Part (c)(ii)

Most learners gained the mark for recognising that estimating p from the observed data led to a second restriction.

Question 5

This question was generally well answered, with most learners recognising the need to calculate the mean and variance from a Negative Binomial distribution and then apply the central limit theorem.

Almost all responses gained the first two marks for finding the mean and variance, and many then applied their mean successfully to the normal distribution, with stronger responses using the required notation for sample means. Many responses stated the standard deviation rather than the variance in the Normal model, possibly because the calculator requires the standard deviation when finding a probability; correct use of the variance nonetheless recovered the M1 for many learners. Responses that did not gain marks, despite recognising that this was a central limit theorem question, did so because they squared or square-rooted the wrong quantity for the variance or standard deviation, or applied a continuity correction when calculating the probability for the dependent M mark.

Question 6

Part (a)

This part was well answered, with most learners able to write down $G_x(1) = 1$ and use this to form the correct unsimplified equation in k , leading convincingly to $k = 27$.

Part (b)(i)

Responses to this part were mixed. Stronger responses either rewrote $G_x(t)$ in a form suitable for a binomial expansion or used the equivalent differentiation approach; both routes were seen. A number of learners instead attempted to expand the denominator directly without first extracting the required factor, which incorrectly sometimes incorrectly led to the answer of $27/64$. A small number of learners recognised this probability distribution as $X + 1 \sim \text{NB}(3, 0.75)$ and made good progress here and in part (ii).

Part (b)(ii)

This part was less challenging than part (b)(i). A mix of the two methods was seen: some learners continued the binomial expansion to find the further coefficients required, while others used the equivalent differentiation approach; both were credited where used consistently and accurately.

Part (c)(i)

This part was answered well, with some learners making efficient use of the differentiation they had already carried out in part (b)(i) rather than starting afresh. As this was a 'show that' question, all stages of working were required to gain full credit; a number of otherwise correct attempts omitted the final substitution needed to demonstrate the given answer.

Part (c)(ii)

There were many good attempts at the further differentiation required for this part, with many learners going on to produce fully correct responses.

Part (d)

Most learners recognised the need to multiply through by t^6 , but a number struggled when replacing every instance of t with t^2 in the resulting expression, which prevented them from reaching the required simplified form.

Question 7

Part (a)

Learners were generally well versed in what a Type 1 error was, and very few responses did not identify that the probability of a Type 1 error was the significance level. Most responses gave the expected answer of 0.05, though a few equivalent forms such as 5% were seen. A number of responses attempted a calculation leading to a probability of around 4%, which suggested that some difficulties with the topic of errors and with the requirement to state rather than calculate the answer.

Part (b)

This part proved to be more challenging with following a clear structure in answering the question.

Responses that engaged well with finding a Type 2 error sometimes struggled to adapt to the unknown sample size when finding the critical value, though many were able to persevere to the simultaneous equations needed for a solution. In some responses, learners either treated μ as fixed at 42 in their attempt at a critical value or confused the roles of 42 and 45 in the question. These responses showed some difficulties working with the given distribution $X \sim N(\mu, 8^2)$, which required standardising $\bar{X} \sim N\left(\mu, \frac{8^2}{n}\right)$; incorrect variances were seen as a result. Responses that engaged with the requirements of this part were given credit for attempting a standardisation in terms of n , which was the first accessible mark. Learners who progressed to a critical value usually found $z = \pm 1.6449$, consistent with the 5% tail, though a number of responses gave a critical region rather than a critical value. Some responses also used $z = 1.2816$, consistent with 0.9, in place of the value needed for the 5% tail.

The z value for a probability of 0.9 was used incorrectly rather than the value for 0.05, and many responses did not reach the point of comparing their standardised critical value with 1.2816. Responses that did reach this point were usually successful in obtaining $n > 60.8$ and hence $n = 61$. It was not unusual to see -1.2816 used instead of the correct 1.2816, leading to an implausible value for n ; few of these responses went on to reconsider whether a sign error might explain this. Learners are reminded of the value of drawing a diagram to check the sign of a z value.

In a small number of otherwise correct responses, marks were not awarded at the final stage through giving a range rather than a minimum value, or through truncating z values, which led to the accuracy not being awarded.

Part (c)

This part was answered well, with most learners recognising that increasing the sample size would increase the power of the test. Some responses included comments about the standard deviation or variance, suggesting some uncertainty about the requirement of the command word 'state'. Responses indicating that the power would decrease were rare, and most learners gained this mark even where part (b) had been minimal or unsuccessful.